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‘Fresnel aether drag’ in a transversely moving medium

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The transverse ‘Fresnel aether drag’ experienced when light passes through a refracting medium moving at right angles to the original direction of the light, and confirmed indirectly by Airy’s water-filled telescope experiment, has been observed directly. In principle a beam of light is projected through a rotating disk of glass, and the consequent transverse displacement of the beam is measured by using a photoelectric alinement technique in which one grid is imaged on another. The main difficulties in the experiment arise from displacements of the light beam due to other causes: optical inhomogeneities in the disk, skewness in its mounting, and changes in its figure due to centrifugal force. These disturbances are reduced by more than a thousand-fold by making the light traverse the disk twice, by using two observing systems at opposite ends of a diameter, and by an optical correction device for inhomogeneities analogous to the corrector bar on a lead screw.

For light traversing a disk of 2.465 cm thickness and refractive index 1.524 at an operating radius of 13.75 cm, when the disk is reversed from +1501.9 to –1501.9 rev/min the displacement observed is $6.175 \text{ nm} \pm 0.016 \text{ nm}$ standard deviation. The displacement (to be expected from the Fresnel aether drag formula) is 6.174 nm.

Arago’s observation (1810) that the apparent refraction in a moving prism is equal to the absolute refraction in a fixed prism led Fresnel to a remarkable result (1818) in the theory of a luminiferous aether. Accepting Young’s demonstration (1803) of the wave character of light—with its consequence that the velocity of light is inversely proportional to the refractive index of the medium—and following Young’s suggestion that this index might depend on the concentration of aether within the medium, Fresnel postulated (by analogy with the formula relating velocity and density in the propagation of sound) that the concentration should be proportional to the square of the refractive index. From a model based on this concept Fresnel concluded that light travelling in a medium of refractive index n that was itself in motion with velocity v relative to the observer would be ‘dragged’ by the medium so as to appear to have an additional component of velocity $v(1 - n^{-2})$ in the direction of motion of the medium.

Fresnel’s prediction was experimentally verified for the cases where the light and the medium are moving in the same or opposite directions by Fizeau (1859), who observed the effect of allowing water to flow suitably along the light paths of an interferometer; his experiment was repeated more accurately by Michelson & Morley (1886). And although Fresnel had depended on the notion of an aether in deriving his formula, von Laue (1907) showed that relativistic theory gave the same result to the first order in v/c ; it is therefore still permissible to think pictorially of ‘aether drag’ in experiments designed to investigate its first-order effects.

There are, incidentally, non-relativistic corrections to Fresnel’s formula which have to be applied in some instances. If, as in Fizeau’s experiment, the medium is

moving directly towards or away from the observer (i.e. directly with or against the light), the wavelength of the light in the medium will be different (by the Doppler effect) from what it would have been if the medium had been at rest; and so, if the medium is dispersive, the refractive index to be used in the formula has to be that appropriate to the modified wavelength. This correction, which gives for the aether drag in the Fizeau experiment $v(1 - n^{-2} + \nu \, dn/n \, d\nu)$, where ν is the frequency in the observer's frame of reference, was first pointed out by Lorentz (1895) and verified by Zeeman (1915). Zeeman (1920) also obtained results for solid media by shuttling glass or quartz rods to and fro along the light paths of an interferometer. In this instance the Doppler correction is modified by the motion of the refracting interface in the line of sight, and the expression for drag then becomes $v(1 - n^{-2} + \nu \, dn/n^2 \, d\nu)$, as discussed by Landsberg (1961) and Evans (1969).

In contrast to the longitudinal aether drag, the transverse effect does not seem to have been observed directly. It was, of course, inferred from Airy's null experiment (1872) in which he showed that the stellar aberration constant as measured with a water-filled telescope was the same, to within the accuracy (a few parts per cent) of his observations, as that given by air-filled instruments (20".496). Lodge (1893) considered the possibility of making a direct observation by projecting a beam of light through a rotating glass disk, but he concluded that the experiment was impracticable because he would have required a glass cylinder 3 m long and 1 m in diameter, rotating at 3000 rev/min, to give a transverse displacement of 1 μm , the smallest shift then optically detectable. Later developments (see, for example, Jones 1961) have made the experiment feasible; and since it is always desirable to observe a physical phenomenon directly when possible, an experiment to verify the Fresnel formula for the transverse case has now been performed. It is described in this paper.

PRINCIPLE OF THE EXPERIMENT

Expected magnitude of the aether drag effect

If a transparent disk of thickness t and refractive index n is spinning about its principal axis at ω radians per second, and a beam of light passes through the disk parallel to the axis of rotation and at a distance r of this axis, then the beam will undergo a tangential displacement δ due to aether drag such that, to the first order,

$$\delta = r\omega t(n - n^{-1})/c.$$

For a disk where $r = 15$ cm, $t = 2.5$ cm, and $\omega = 130$ rad s⁻¹, δ is of order 1 nm. The sensitivity of optical alignment devices is currently ample for detecting such a displacement: any difficulty in the experiment therefore arises not in achieving the necessary sensitivity but in reducing the disturbing effects of the much larger displacements that arise from other causes. We now discuss the most serious of these effects.

Effect of skewness in disk mounting

Supposing that the disk is ideally uniform in thickness and in refractive index, there will inevitably be a sideways displacement of the light beam whenever the plane of the disk and the direction of the light are not perfectly normal to one another. If the angle between the beam and the normal to the disk be θ then the sideways displacement of the light in passage through the disk will be $\alpha t [1 - (1 - \alpha^2)^{\frac{1}{2}} / (n^2 - \alpha^2)^{\frac{1}{2}}]$, where $\alpha = \sin \theta$. For small values of θ , such as those with which we shall be concerned, this reduces to $\theta t(1 - n^{-1})$; it may be conveniently termed the ‘optical micrometer displacement’, for it is commonly utilized in the standard ‘optical micrometer’ device for generating small optical displacements by rotation of a transparent plate.

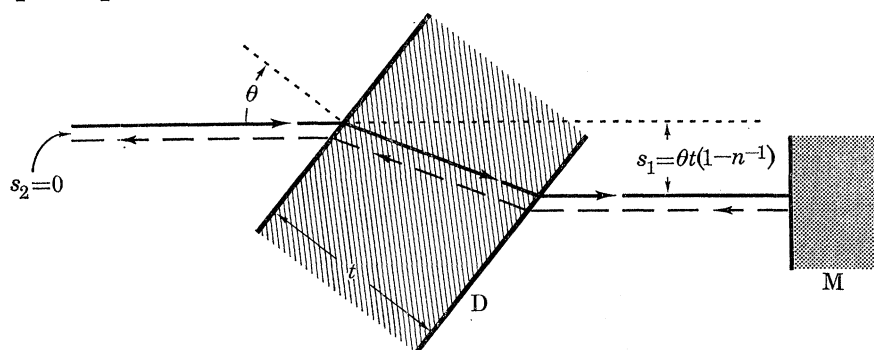


FIGURE 1. The transverse displacement s_1 of a beam of light passing through a plane-parallel block of medium D is cancelled (s_2) by reflecting the beam in the mirror M back along its original path.

Even in the relatively simple operation of mounting the disk on the shaft on which it is to be rotated, it is not easy to set it so precisely that its normal coincides with the shaft axis to better than $10''$ or 5×10^{-5} rad. If a 2.5 cm thick disk is skewly set by this amount, it will deflect the beam by about 400 nm, or some hundred times the entire displacement that we are seeking to observe. And, even if the disk is set perfectly on the shaft, the axis of rotation of the latter may change with speed because of the play in the bearings necessary to permit easy rotation; we therefore need an optical system which automatically eliminates this type of displacement, at least to the first order. Fortunately, it is possible to devise a system that fulfils this purpose and at the same time doubles the genuine ‘aether drag’ effect. It is effected by reflecting the light back again through the disk, as shown in principle in figure 1. After its first passage through the parallel slab of medium D the beam is displaced laterally by the optical micrometer effect; but if it now strikes the mirror M normally, it returns along its previous path so that the resultant lateral displacement of the beam after retraversing the slab is zero. This is true for all angles for inclination of the slab, and so accidental displacements due to skewness are eliminated. At the same time, if the medium in the slab is moving transversely the aether drag will

operate on the light in the same direction on both the original and return passages, thus doubling the effect. In the experiment as detailed in the next section, the application of this principle is slightly complicated by the presence of lenses between the slab and the mirror, but the only necessary modification is to replace the plane mirror by a corner reflector.

Optical system: principles

Light from the source S (figure 2) is focused by the condenser lens L_1 through the disk D and the lens L_2 , forming an image of S mid-way between the two mirrors M_1 and M_2 that together constitute a corner reflector. The light then comes back on a parallel path through lens L_3 (similar to L_2) and again through D to L_4 (similar to L_1). A grid G_1 is interposed between L_1 and L_2 in the principal focal plane (allowing for the presence of D) of L_2 ; a unit magnification image of G_1 is therefore formed in the plane G_2 , where an identical grid is placed. Regarding, now, a point g_1 on the axis of L_2 and in the plane of G_1 , light diverges from it through D towards L_2 , whence

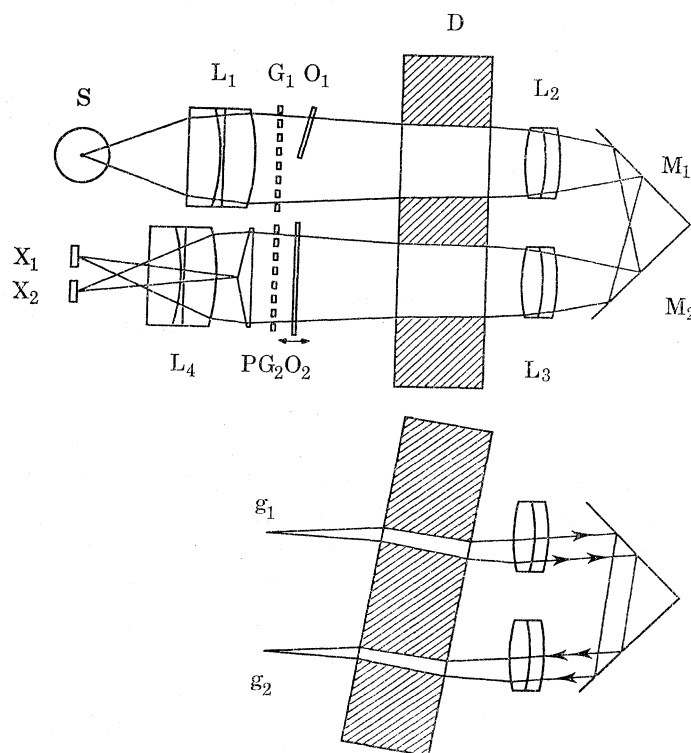


FIGURE 2. Scheme of an optical system using the principle of figure 1, as applied in the aether-drag experiment.

it emerges as a parallel beam; it is then reflected back by M_1 and M_2 through L_3 , where it begins to converge to g_2 on G_2 . If D is skewly set, as in the lower diagram of figure 2, the diverging beam from g_1 is displaced sideways, and it strikes L_2 in such a way that, although it still emerges as a parallel beam, its direction is no longer

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parallel to the optic axis of L_2 . It returns parallel to its new direction after reflexion in M_1 and M_2 , and it is therefore made by L_3 to converge towards a point in G_2 that is displaced from the optical axis; but the second passage through D displaces the beam back again so as to compensate almost exactly for the displacement caused by the first passage. The beam therefore comes to the same focus g_2 on the optic axis of L_3 in the plane G_2 , whatever the skewness of D to this axis; a similar argument applies to all other points in the image.

Although the position of the images of G_1 in the plane of G_2 is therefore made independent of the skewness of D (to the first order: there may be a second order non-cancelling effect since when D tilts it recedes from L_3 as it approaches L_2 and vice versa), any sideways ‘drag’ on the light due to the transverse motion of D will be in the same direction for each passage through the disk. The image of G_1 will therefore then be displaced sideways by twice the amount due to a single passage.

Thus for a 25 mm thick disk, the ‘aether drag’ effect will be that due to passage through 50 mm of medium. Tests of the optical compensation system as used in the present experiment show that the residual optical micrometer effect after the double passage through the disk is 150 times less than it is for a single passage. A factor of 300 in discrimination of aether drag against skewness effects is therefore gained by the double passage.

System for displacement detection

In principle, if the image of G_1 falls in the plane of G_2 , any transverse displacement of this image will cause more or less of the light that passed through the spaces in G_1 to be intercepted by the bars of G_2 . Since the grid interval is about 0.2 mm and the displacement that we wish to measure to 1 % is about 1 nm, it follows that we have to detect a change of one part in 10^7 in the amount of light passing through G_2 . A convenient way of doing this is to interpose a thin transparent parallel plate O_1 obliquely across one half of G_1 so that the light from this half is displaced sideways (by the optical micrometer effect in O_1) to such an extent that if it was originally passing exactly through the spaces in G_2 it is now intercepted exactly by the bars. Viewed from behind G_2 , one half of the field remains bright while the other half becomes dark. If another plate O_2 is now obliquely set at a suitable angle (half that of O_1 if the two plates have the same thickness) across the whole front of G_2 , the effect is to side-step both halves of the image of G_1 to such an extent that 50 % of the light in each half passes through G_2 ; but any further sideways displacement of this image (as may be expected from the aether drag) will cause one half to brighten while the other half darkens, and a reverse movement will cause a reverse effect. If the light from the two halves is deflected suitably by a biprism P and focused by a condenser lens L_4 (similar to L_1) the result will be two separated images of the source S of equal intensity when the system is in balance. Photocells X_1 and X_2 are placed at these image points and connected in a bridge system; any movement of the image of G_1 then upsets the balance by making more light fall on one photocell than the other, and the displacement can thus be detected.

Effect of rotation on the figure of the disk

A preliminary experiment using the system as described above (Jones 1971) revealed an unforeseen difficulty. It had been expected that the birefringence induced by the centrifugal stress might become appreciable, but in practice this effect was small enough to be neglected at the relatively low angular speeds necessary to give a detectable aether drag. However, even at these speeds, it was found that there was a large displacement of the image of G_1 relative to G_2 which was in the same direction irrespective of the sense of rotation of the disk, and proportional to the square of the rotational speed. It was traced to the change in figure of the disk as it rotated; owing to centrifugal force, the originally parallel-plane disk became concave. The beam of light passing through it therefore suffered not only the aether drag tangentially, but also a much larger deflexion radially outwards because it was in effect passing through a prismatic section in the radial plane; and this latter deflexion, like the aether drag, was augmented by the second passage through the disk.

The rotational effect showed up strongly in the preliminary experiment because the grids G_1 and G_2 had been set parallel to one another, and therefore not with their bars radial to the disk. The resultant outward movement of the image of G_1 in a radial direction thus had an appreciable component across the lines of G_2 , and this component did not change sense when the rotation was reversed.

It was an unforeseen merit of the optical arrangement previously described that when G_1 is rotated in its own plane its image in the plane of G_2 rotates by an equal and opposite amount, and so each grid can be set with its bars parallel to that radius of the disk which passes through the axis of its optical system, while at the same time the bars in the *image* of G_1 are parallel to the bars of G_2 . When this change was effected, the system could be made insensitive to the radial component of deflexion caused by the increasing concavity of the disk with speed.

The magnitude of the radial deflexion of a beam of light on passing through the disk can be estimated from a formula for the distortion of the disk provided by Professor A. Green for the distortion of the disk with rotation. If a uniform disk of radius a and thickness t (when at rest) rotates about its axis at angular speed ω rad s⁻¹, the thickness z at a radius r from this axis becomes

$$z = t[1 - \rho\eta\omega^2\{(3 + \eta)a^2 - 2(1 + \eta)r^2\}/4E],$$

where ρ represents the density η the Poisson ratio, and E the Young modulus, for the disk material, giving

$$dz/dr = \rho\omega^2 r t \eta (1 + \eta) / E.$$

Since, for glass, $\rho = 2.60$ g cm⁻³, $\eta = 0.25$, $E = 7 \times 10^{11}$ dyn cm⁻² (7×10^{10} Pa) at $r = 13.7$ cm and $\omega = 50\pi$ rad s⁻¹.

$$dz/dr \approx 9 \times 10^{-7} \text{ rad.}$$

The light will therefore suffer a radial deflexion due in effect to passing through

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a prism of this angle; and for glass of refractive index 1.5, the deflexion will be about 5×10^{-7} rad. This will be increased to 10^{-6} rad for the double passage; and the total displacement of the image of G_2 will be this angle multiplied by the focal length of L_2 or L_3 (7.5 cm), or 7.5×10^{-6} cm.

The foregoing estimate ignores the effect of the central hole (80 mm diameter) in the disk, but this is likely to be small in the peripheral region of the disk used in the experiment; and the estimate agrees well with the magnitude of effect observed. The undesired radial displacement is therefore at least ten times greater than the expected tangential displacement due to aether drag; and the grids G_1 and G_2 must therefore be carefully set with their lines parallel to the appropriate radii of the disk if the effect of the undesired displacement is to be reduced to a few parts per cent of the aether drag displacement. Any remaining component of the radial effect can then be eliminated by observing the overall deflexions with the disk running at the same speed in clockwise and counterclockwise directions, and taking the difference.

Optical system: details

Since the foregoing considerations have determined the main principles of the optical measuring system, it is convenient now to describe the physical details, which are shown in figure 3. The several components are mounted on a base of duralumin 350 mm $\times$ 101 mm $\times$ 30 mm, which is sufficiently rigid to maintain them in the desired geometrical relation (i.e. their relative positions changing by not more than, perhaps, 1 nm over a period of 1 h). The source S is a 4 V 0.8 A krypton-filled tungsten lamp (similar to Ediswan type MES 40 8 K, but specially made with a more

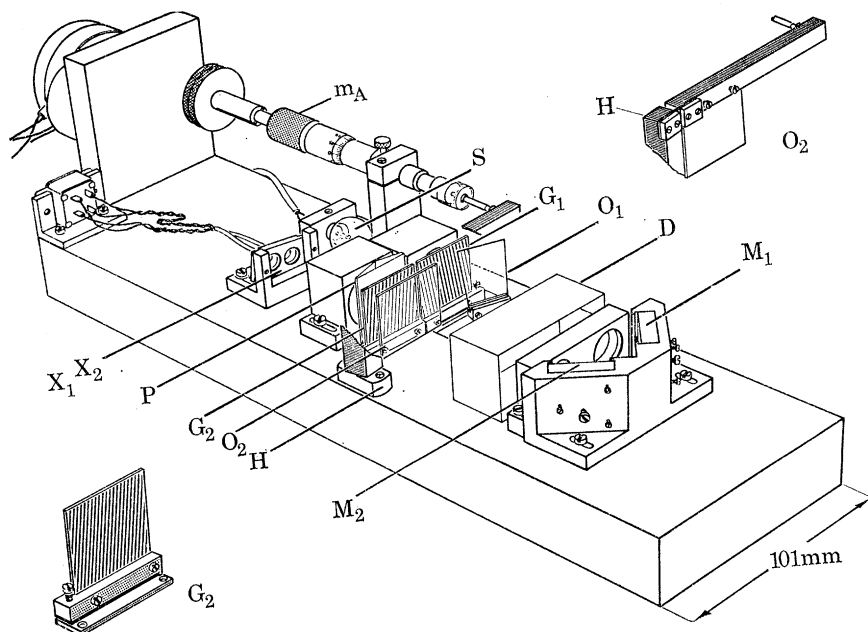


FIGURE 3. Sketch of optical system using the scheme of figure 2.

compact filament) operated at 3.5 V. The condenser lens L_1 is a pair of achromats each of 25 mm diameter and 54 mm focal length placed nearly in contact. The grid G_1 is 25 mm square, with its grid lines inclined at an angle of about 6° to the normal to the base, so that the centre line of the grid will converge with the correspondingly inclined line of G_2 to a point that lies on the axis of rotation of the aether-drag disk. The grid interval (55 % bar plus 45 % space) is 0.17 mm. A circular aperture of 20 mm diameter in the plane of G_1 limits the diameter of the light beam so that all the light passing through G_1 will ultimately fall comfortably inside the entrance pupil of L_4 . The glass block D shown in the figure is of the same thickness and material as the disk, and is employed in setting up and checking the optical system before the latter is incorporated in the main assembly; the cross-section of the light beam where it passes through the disk is roughly a circle of 10 mm diameter. Lens L_2 is an achromat of 17 mm diameter and 75 mm focal length. The corner reflector system M_1M_2 (shown as independently adjustable mirrors in the figure, but replaced in later experiments by a single aluminized right-angled prism) reflects the light back along a parallel path separated from the original path by 28 mm. This separation is made as small as possible, but it cannot be less than the distance between the centres of L_1 and L_4 in their mounts; an autocollimating system would have been better in this respect, but it would have been less efficient optically.

Lens L_3 is similar to lens L_2 ; and when the system is in correct adjustment the cross-section of the beam returning through D should again be roughly a circle of 10 mm diameter. The optical micrometer plate O_2 is of glass of refractive index 1.516 and is 1.873 mm thick; it is carried on an arm (largely cut away in the figure) which pivots on a leaf-spring hinge anchored to the base at H and actuated by the mechanical micrometer m_A . G_2 is similar to G_1 and, as is shown exaggerated in the figure, is G_1 's mirror image. The biprism P is of Perspex and of such an angle as to separate the images of S sufficiently (about 10 mm) in the plane X_1X_2 to permit two silicon photodiodes (20th Century type OSD 50) to be mounted side by side. In later experiments, incidentally all components of the system after G_2 have been replaced by a pair of large silicon photocells (Ferranti type MS 2) placed side by side in the position previously occupied by the biprism P.

The use of a double measuring system

When the optical system was first set up, grids G_1 and G_2 had been set with their lines parallel to one another, as described earlier. When they were re-set to converge so as to eliminate the effect of radial deflexion of the light by the changing figure of the rotating disk, this introduced a new difficulty. With the two grids parallel, the system had been insensitive to a tilt of the disk about an axis that was perpendicular to the light path and to the lines of the grids, because the effect was merely to move the image of G_1 slightly upwards or downwards in a direction parallel to the lines of G_2 . Such a movement produced no change in the light passing through the grids; but this condition no longer held when the grids were set to converge as indicated in figure 3, as may be seen by considering the effect of rotating the block D

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in the figure about its longest axis. The resultant motion of the image of G_1 relative to G_2 (because of the optical micrometer effect in passage through D) is then no longer directly along the lines of G_2 , but has a component across them, thus causing more or less light to pass through. Thus, when the block D is replaced by the disk, any tilt of the disk that is due either to imperfect mounting on its axis or to drift in this axis with speed, may cause a movement of G_1 that will give a false indication. We have therefore made the system insensitive to the change in figure of the disk with rotation at the expense of re-introducing an undesired component of sensitivity in the system to wandering of the disk axis. Fortunately this, too, can be effectively eliminated by observing simultaneously with two systems, mirror images of one another, at opposite ends of a diameter of the disk (figure 4), where any rotation of the disk about a vertical axis will cause equal and opposite effects in the two systems if the light beams are arranged to travel as shown in the figure. The signals from the two

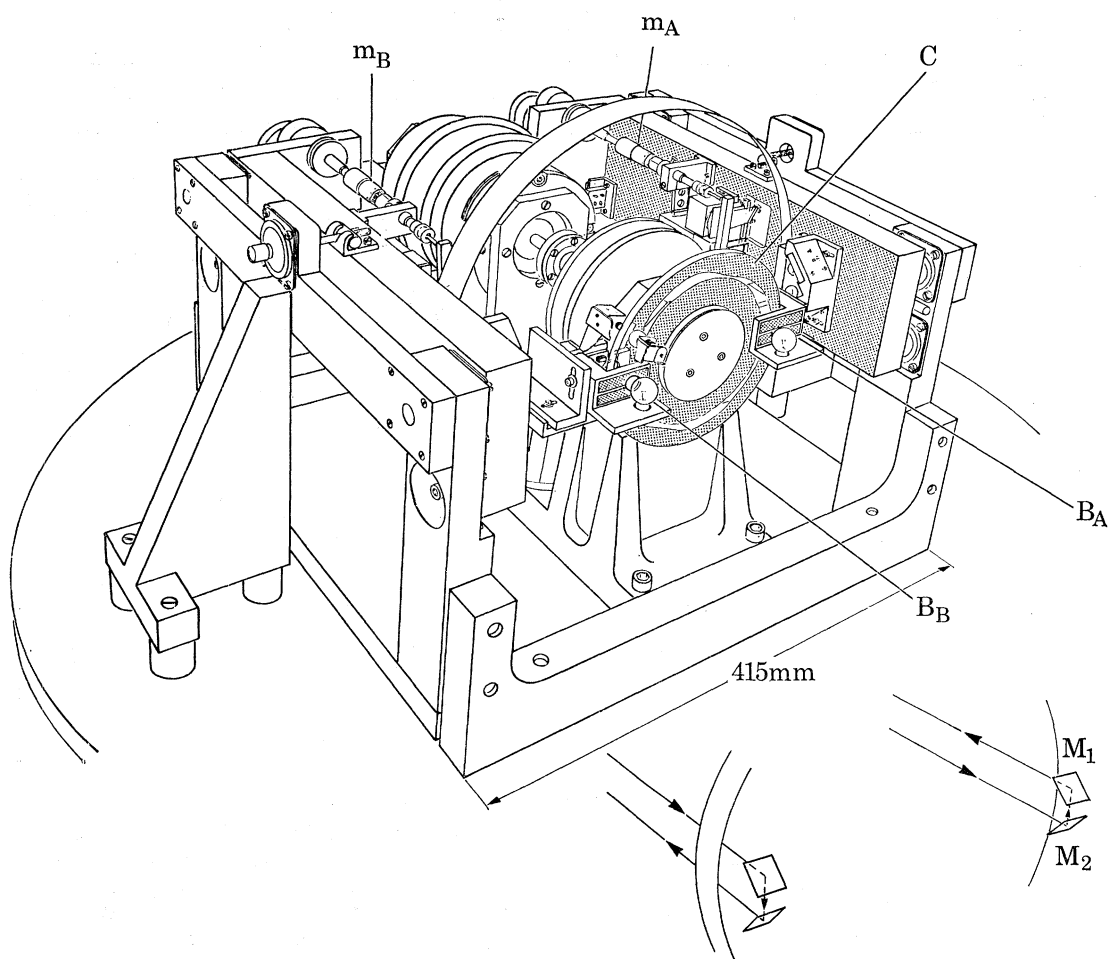


FIGURE 4. Sketch of complete apparatus using two optical systems.

systems can then be combined in such a way as to add the effects of the aether-drag displacement while cancelling those due to disk skewness.

Corrections for disk inhomogeneity

Although the disk (300 mm diameter, 24.65 mm thick, of Chance 524589 glass, by Optical Works Ltd, Ealing) is made from Schlieren quality glass, inhomogeneities in this material can easily be detected by the measuring systems as the disk is slowly rotated. It is difficult to obtain glass with variations in refractive index that are less than about one part in 10^5 ; and even a transverse refractive index gradient of 10^{-6} cm^{-1} would deviate a light ray by $2.5 \times 10^{-6} \text{ rad}$ on a single passage through the disk; this deviation would produce a displacement of nearly $2 \times 10^{-5} \text{ cm}$ in the focal plane of a 7.5 cm lens such as L_2 or L_3 , or at least 30 times the entire deflexion that we are trying to observe. The deviation will not in general be cancelled on the return passage, because the light comes back through a different part of the disk; and variations in the thickness of the disk will cause a similar trouble.

Although these inhomogeneities can be averaged when the disk is rotated, and are therefore inconsequential at the higher speeds of rotation, they produce large deflexions at lower speeds and thus make the mean readings at these speeds difficult

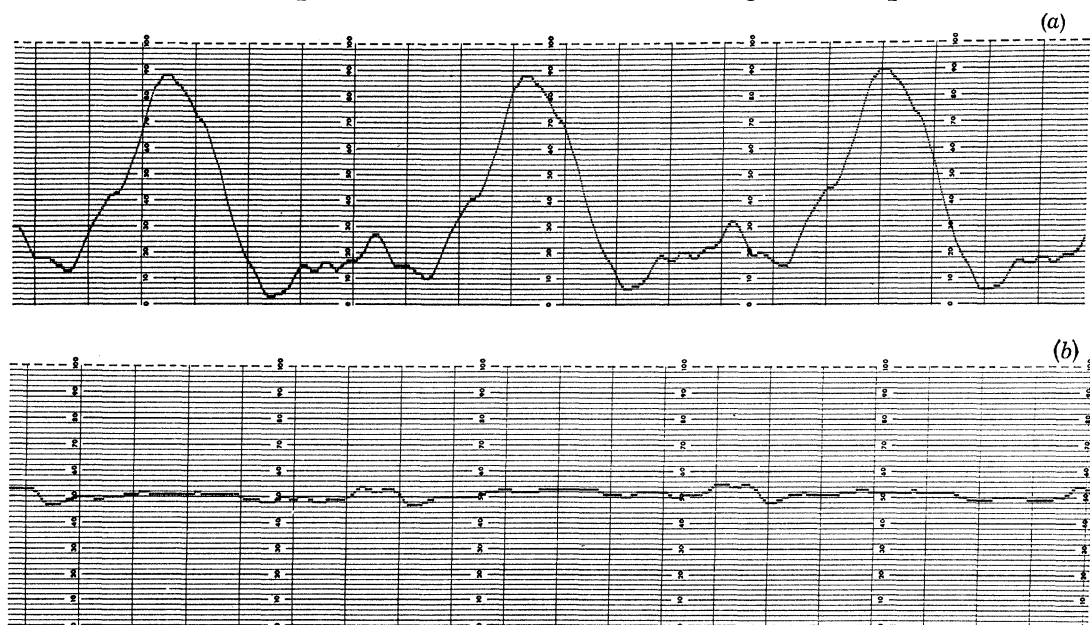


FIGURE 5. (a) Record of zero 'wandering' of one of the optical systems as the aether-drag disk rotates slowly (about 1 revolution in 5 s), due to deviation of the light by variations in the refractive index and thickness of the disk. The record covers nearly three complete rotations, and the total variation corresponds to 320 nm movement of the image of grid G_1 across G_2 , or about 50 times the aether-drag displacement. (b) Corrected record, on the same scale as (a), produced by adding the signal from the photoelectric reading head on corrector disk C to the output of the main photocells. The extreme variation is now reduced to about 22 nm.

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to observe. They can, however, be corrected by the following device. Each measuring system is first used to provide a record (figure 5*a*, which shows that the magnitude of the deviation has been correctly estimated in the preceding paragraph) of the deflexions of the light passing through the disk when it is rotated slowly. The records for the two systems should be identical in pattern, but separated in phase by 180° ; in practice, the patterns may differ slightly, and an average is taken from the two. From the averaged record, a subsidiary disk is made in transparent Perspex and is painted black, except for a track whose width is made proportional to the deflexion of the light on the record. This subsidiary disk (C, figure 4) is then mounted on the same shaft as the main disk, and the transparent track is monitored by a subsidiary lamp and photocell; the output of the latter is fed, along with that of the main photocells X_1 and X_2 , into the measuring amplifier in such a sense and with such a magnitude and phase as to cancel the signal due to the disk errors; each system is corrected separately, with its individual monitoring head taking the signal off the corrector disk in the appropriate phase; the heads are shown at B_A and B_B on figure 4. Figure 5*b* shows the output of one of the measuring amplifiers on the same scale as figure 5*a*, with the correction signal added, showing that the errors have been reduced to less than one-tenth of their original value.

Miscellaneous precautions

Besides the faults that arise directly in the optical system itself, and which have been treated in the previous sections, there are other sources of disturbance which must be reduced before the desired effect can be satisfactorily observed. These sources are principally vibration, temperature changes, and refraction by any gas that is present in the optical path.

Vibration will cause the various components of the optical systems to move relative to one another; some degree of insulation is obtained by mounting the bases of the systems on anti-vibration bushes. Vibration arises mainly from out-of-balance mass in the rotating system or from faulty bearings. The rotating system consists of two parts, the disk assembly and the driving motor, which are joined together by a flexible coupling. The disk assembly and the motor are balanced separately; owing to their axial symmetry it is sufficient to balance them statically in their bearings by a procedure such as is used in standard gyroscope practice. Trials showed that it is possible to obtain ball races of adequate quality to restrict vibration to an acceptable value, and also to maintain the axis of rotation sufficiently accurately for the experiments to be performed with either sense of rotation. These bearings are type 203 FF 3 G-6, manufactured by the Barden Corporation, Danbury, Connecticut, and are simple deep-groove ball races, axially loaded with a force of about 7 kgf to remove end play, and accepting a shaft of 17 mm diameter. This shaft is sufficiently short (50 mm) and of increased diameter (80 mm) between the bearings to avoid any detectable 'whip'.

Temperature variations in the apparatus will result in changes in the relative positions of the various optical components, due to linear expansion, and they can

also cause refractive index variations in the disk. If, during the course of a run, the temperature of the Duralumin base in the region of the grids G_1 and G_2 changes by 0.01 K relative to the region of the base supporting the mirrors M_1 and M_2 , this will cause a zero shift in the optical system of the same magnitude as the expected image shift due to aether drag. The most serious cause of temperature change is the heat dissipated in the driving motor, and this is minimized by water cooling; in addition, radiation screens are placed between the motor and the optical systems, and also around the disk bearings, which are a detectable source of heat.

The experiment has to be performed in an evacuated chamber because of the variations in refraction due to gas in the optical paths. These variations are due to density gradients associated either with convection currents or with aerodynamic drag by the rotating disk. They are serious enough to obscure the aether drag effect unless the pressure is substantially reduced, and even at 1 mmHg pressure they produce marked effects which do not change sense with the direction of rotation, and which are presumably due to a centrifugal effect in the gas. These troubles can be effectively eliminated by working at 1 μ mHg pressure or lower.

Troubles arising in the electronic systems due, for example, to pick-up from the various electric motors used in the apparatus, can be eliminated by relatively simple screening and earthing, since the demands of the experiment on the sensitivity of the photoelectric systems are not very stringent; typically the greatest sensitivity required is at least twenty times less than the ultimate limit of the photocells.

Besides amplifying the signals from the photocells, the electronic systems also serve to smooth out the residual disturbances due to imperfections in the disk and its mounting. In fact, the methods of compensation previously described may be regarded as cumulative steps to reduce the residual errors (by a thousand-fold or more) to the stage where they are small enough not to overload the amplifiers. And since they are cyclic they can then be averaged out, except at the very lowest speeds, with an amplifier time constant of 2 s.

EXPERIMENTAL PROCEDURE

The two optical systems are set up as identically as possible, starting with the selection of lamps that have filaments of the same shape, and proceeding to the adjustment of the optical components so that the light beams have the same cross-sections at corresponding points in the two systems. This is particularly important where the light beams pass through the disk. The grid inclinations are set in each system by an optical template cut to the angle calculated to make the centre lines of the grids converge to the axis of rotation of the disk; the template is a simple straight edge mounted at the correct angle to the base, and each grid is mounted on a spring hinge that can be controlled by a set-screw to bring the grid lines parallel to the straight edge. By observing with a long focus microscope the grids are set to within a few minutes of arc of the correct direction.

All components are matched as well as possible in the two systems; the photocells

have also to be matched in pairs for the individual systems. Even so, and with matched amplifiers, the two systems are not likely at this stage to have overall sensitivities that are matched to better than 10 %. The sensitivities of the two systems are now brought within 1 % of one another by adjusting the lamp current in one system. The sensitivities are measured by producing a calibrated shift in the image G_1 relative to G_2 by rotating the optical micrometer plate O_2 by means of the mechanical micrometer. This rotation is in turn calibrated by attaching a mirror to O_2 and using an autocollimator. The linear shift of the light passing through the optical micrometer plate is then calculated from the measured thickness (1.873 mm) and refractive index (1.516) of the plate.

The next step is to mount both systems so that they are each correctly positioned with respect to the disk, with their grid lines converging towards the axis of rotation. The signals from the optical corrector disk are then coupled in to the appropriate amplifiers. The phase of the correction signal for each system is set by first observing the outputs of the system and of the corrector on a twin recorder, and adjusting the position of the corrector reading head appropriately. The signals from the system and from the corrector are then coupled in opposition, and the lamp current in the corrector adjusted to give the best neutralization.

The output of each corrected system is recorded separately; any change of axis with sense of rotation shows up at this stage, and the information can be used to adjust the disk bearings if the shift of axis is excessive. In all the measuring runs reported in this paper, the shift of axis on reversal has been made small enough to cause a deflexion of less than 5 % of that due to the aether drag at 1500 rev/min. This residual effect is further reduced when the outputs of the two systems are added together in the correct way; this addition also cancels the effect of any residual wobble. Tests have shown that the cancellation reduces by a factor of about 20 the residual effects in the individual systems.

The whole of the apparatus shown in figure 4 is evacuated and a typical series of runs starts with a calibration of the sensitivities of the two measuring systems, both separately and in conjunction. Small motors are made to move the mechanical micrometers m_A and m_B by 100 μ m displacement, and the resultant optical displacements are measured on pen recorders with the associated electronic photocell amplifiers operating at one-fiftieth of the sensitivity that will be used in recording the aether drag; the amplifiers have to be thus attenuated during calibration because otherwise the mechanical micrometer shifts would be too small to be read. The attenuation factor of each amplifier is set to be within 0.1 % of its nominal 50, and this is checked from time to time.

If the disk is stationary, it suffers a slight differential heating in the regions where it is traversed by the light beams. The effect may be due more to the local proximity of the bases of the optical systems (which are themselves slightly warmed by the heat dissipated in the light sources) than to the direct absorption of energy from the light beams. The result is that the optical thickness of the disk is locally altered, and this shows as an additional deflexion on traces such as that of figure 5*b* which slowly

decays as the disk continues to rotate and the temperature distribution becomes uniform. Before any measuring run is attempted, the disk is therefore rotated slowly for several minutes.

A typical run is performed by running the motor up to its synchronous speed (normally 1500 rev/min) as quickly as the inertia of the disk will allow. This takes about 30 s, and the system is allowed to run at full speed for another 40 s; the motor connexions are now switched, so that the disk and motor come to rest in a further 30 s, and then gather speed for the reverse cycle. The whole operation is repeated, giving a record which shows successive deflexions due to the aether drag twice in each direction. Simultaneously a record is made of the disk speed, using the pulses generated by the passage of a slit in the corrector disk C between an auxiliary lamp and photocell. The scales of the two records are matched fairly closely by adjusting the recorder sensitivities; and the speed calibration is checked with a standard electronic timer.

RESULT

A typically good record is reproduced in figure 6. It shows the trace of the optical deflexion as the disk (originally and finally running at about 180 rev/min) is speeded up to a nominal 1500 rev/min and reversed; superimposed in the overlay is the simultaneously recorded trace of the disk speed. From these records a reading is taken of the total deflexion of the light when the speed is changed from +1500 to -1500 rev/min. Thirty-seven such pairs of records (taken over about 4 weeks of observation) have been used, each to evaluate a single value of the deflexion now reported.

After calibration by the optical micrometer, which is also used to confirm that the deflexion is in the direction of rotation of the disk, the values obtained from the individual records in nanometres units (and in order of time) are:

6.02, 6.02, 6.21, 6.02, 6.17, 6.17, 6.07, 6.14, 6.21, 6.28, 6.06, 6.30, 6.36, 6.15, 6.33, 6.20, 6.24, 6.06, 6.18, 6.20, 6.10, 6.26, 6.23, 6.23, 6.41, 6.30, 6.12, 6.08, 6.23, 6.10, 6.06, 6.16, 6.12, 6.16, 6.16, 6.16, 6.16

These give a mean result for the optical deflexion of 6.174 nm with a standard deviation of 0.016 nm.

The value to be expected for the first-order aether drag can be calculated from the following measurements relating to the disk:

thickness	= 2.465 cm
mean operating radius	= 13.75 cm
speed range	= +1501.9 to -1501.9 rev/min
refractive index	= 1.524 (at 589.3 nm).

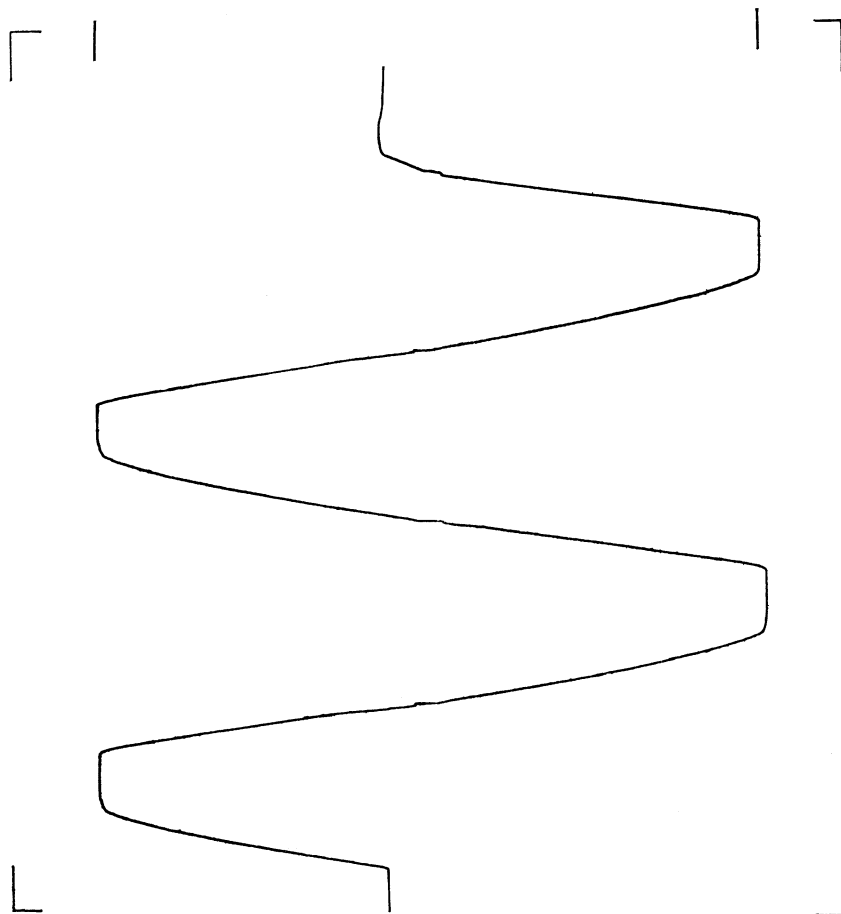
These give an expected value for the deflexion of 6.175 nm for the double passage of the disk.

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– 2165 cm/s
– 1500 rev/min

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+ 2165 cm/s
+ 1500 rev/min



Overlay to figure 6, page 351.

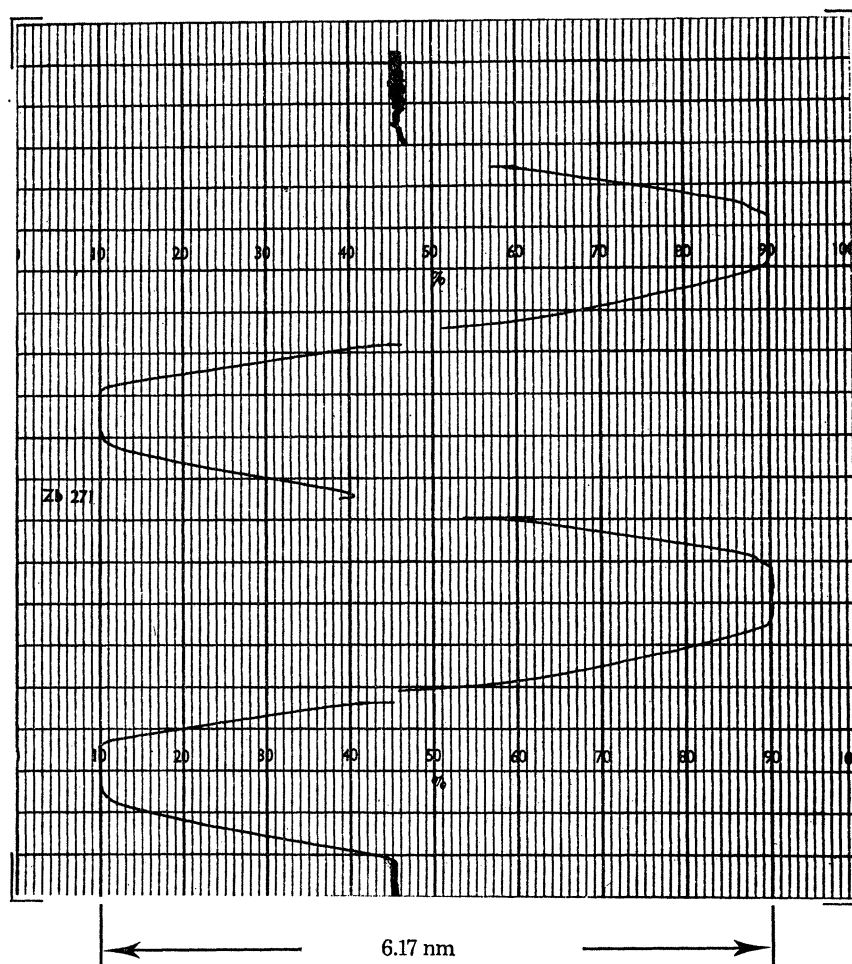


FIGURE 6. Sample record showing (abscissae) the transverse deflexion of light passing through the aether-drag disk, as this is accelerated from about 180 to 1500 rev/min and then reversed to 1500 rev/min in the opposite direction. The overlay shows the simultaneously taken record of the disk speed. Timing divisions (ordinates) are at 20 s intervals, and the time constant of the recording amplifier is 2 s. The gaps in the deflexion trace are due to the recorder pen being lifted while the disk is moving slowly, and the residual effects of optical inhomogeneities and skewness are therefore not being adequately averaged.

The close agreement with the observed value must to some extent be fortuitous: in particular it is difficult to determine the mean operating radius to better than 0.3 %, since this involves a visual determination of the positions of the central axes of the light beams where they pass through the disk. Errors in the other quantities are likely to be smaller; the effective refractive index could be perhaps 0.4 % less than that used in the calculation (since the sensitivity of the photocells rises towards

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– 2165 cm/s
– 1500 rev/min

+ 2165 cm/s
+ 1500 rev/min

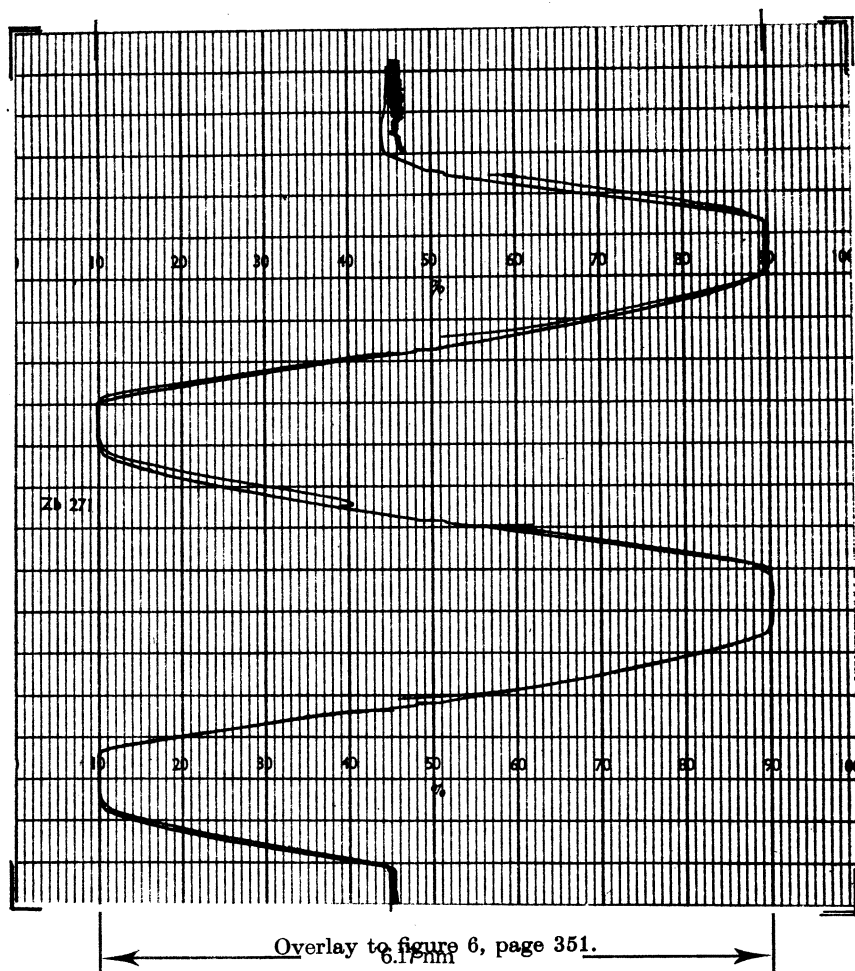


FIGURE 6. Sample record showing (abscissae) the transverse deflection of light passing through the aether-drag disk, as this is accelerated from about 180 to 1500 rev/min and then reversed to 1500 rev/min in the opposite direction. The overlay shows the simultaneously taken record of the disk speed. Timing divisions (ordinates) are at 20 s intervals, and the time constant of the recording amplifier is 2 s. The gaps in the deflection trace are due to the recorder pen being lifted while the disk is moving slowly, and the residual effects of optical inhomogeneities and skewness are therefore not being adequately averaged.

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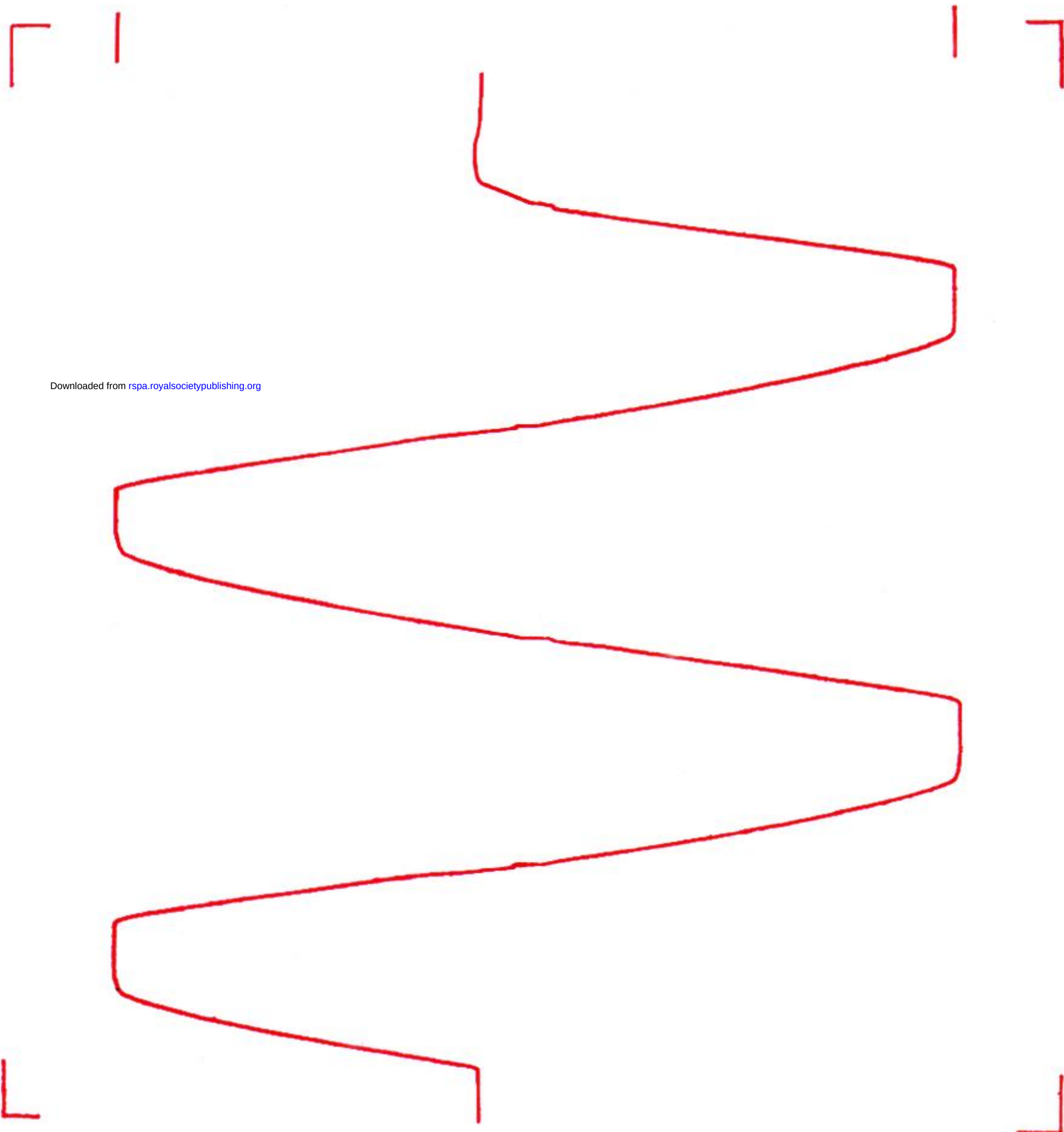
the red) but this will be partly compensated for by the change of refractive index in the optical micrometer plate used in the calibration, so that the error from this source is unlikely to be greater than 0.25 %.

The expected aether drag has been calculated on the assumption that it is the phase velocity in the glass that determines the aether-drag angle. For the particular glass involved, the group velocity would be about 1.5 % less, giving an expected drag correspondingly increased to 6.27 nm. The value actually found is therefore in better agreement with that to be expected from the phase velocity.

I am especially indebted to Dr J. C. S. Richards, who designed the electronics associated with the photoelectric measurements and with the disk speed recorder, to Mr L. Keith and Mr G. Paget for their skilful construction of the mechanical components, and to Mr A. M. Gardner for his continuous assistance throughout the experiment: he took all the records used in the final determination, including that of figure 6. My thanks are also due Mr G. Coull and Mr T. Wratten for preparing the diagrams for this paper, and to Professor A. Green for providing the formula for the disk distortion.

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− 2165 cm/s
− 1500 rev/min

+ 2165 cm/s
+ 1500 rev/min

